



TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CIVICS GROUP

H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

16 September 2022

2 hours

Candidates answer on the Question Paper.

No additional materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and Civics Group in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **ALL** questions.

Section B

Answer **ONE** question.

The use of an approved scientific calculator is expected, where appropriate

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 31
2	/ 19
Section B	
3 or 4	/ 25
Total	/ 75

Section A

Answer all questions in this section.

QUESTION 1

All life on earth is categorized into three domains – Eukarya, Bacteria and Archaea.

- Eukarya: organisms with cells that contain a nucleus as well as membrane-bound organelles
- Bacteria: unicellular prokaryotic cells that do not contain a nucleus and membrane-bound organelles
- Archaea: a unique class of unicellular prokaryotic cells that present many distinctive features from the domain Bacteria

Archaea were first discovered in 1977 in high-temperature environments, such as hydrothermal vents and terrestrial hot springs. They were later also found to be residing in other extreme conditions, such as highly acidic and anaerobic environments.

(a) One feature unique to Archaea is their cell membranes. Fig. 1.1 shows the molecular structure of the membranes of Eukarya and Bacteria and the membranes of Archaea.

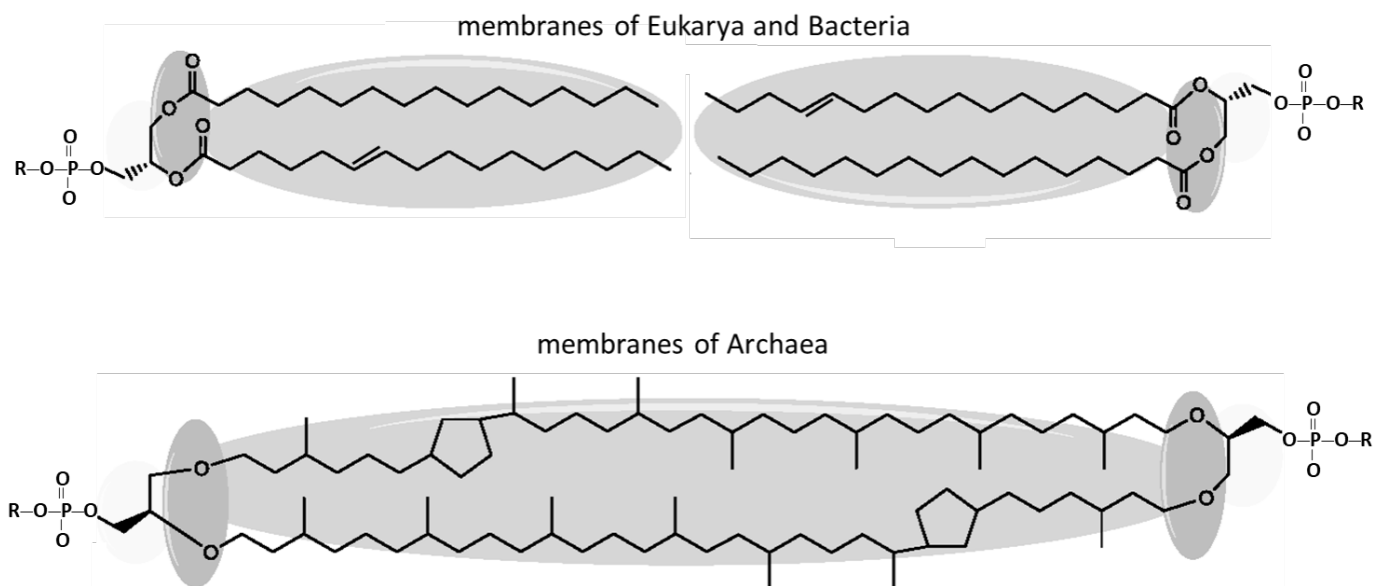


Fig. 1.1

- (i) Describe the structural differences between the membranes of Eukarya and Bacteria and the membranes of Archaea. [3]

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- (ii) Suggest how **one** of the structural differences you have described in (a)(i) enables the Archaea to survive in high-temperature environments. [2]

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- (b) Some archaeal cells synthesize ATP via photophosphorylation, using ATP synthase and a protein called bacteriorhodopsin. Permanently associated with bacteriorhodopsin is a light-harvesting prosthetic group called retinal, which changes conformation upon light absorption. On top of its light-harvesting property, bacteriorhodopsin is also a proton pump.

Fig. 1.2 shows the structure of bacteriorhodopsin and ATP synthase embedded in the archaeal cell surface membrane.

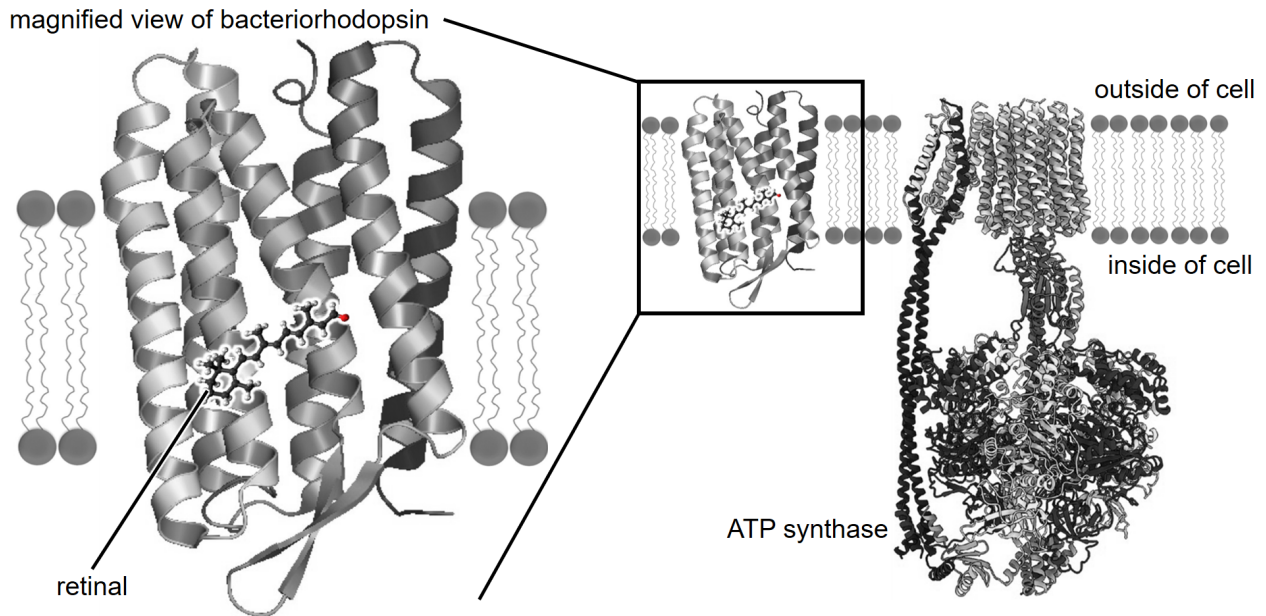


Fig. 1.2

- (i) Outline the structural arrangement of bacteriorhodopsin in the archaeal membrane. [2]

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- (ii) With reference to Fig. 1.2, suggest how photophosphorylation occurs in the Archaea. [4]

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- (c) Before their discovery in 1977, archaeal cells had been wrongly classified as bacterial cells, receiving the name Archaeobacteria, for many years.

Fig. 1.3 shows two possible evolutionary relationships among the three domains of life.

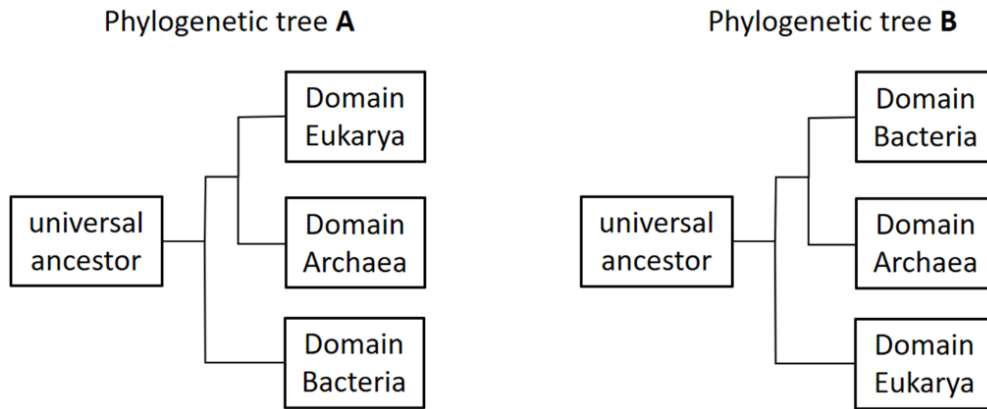


Fig. 1.3

Fig. 1.4 shows the general structure of an archaeal cell.

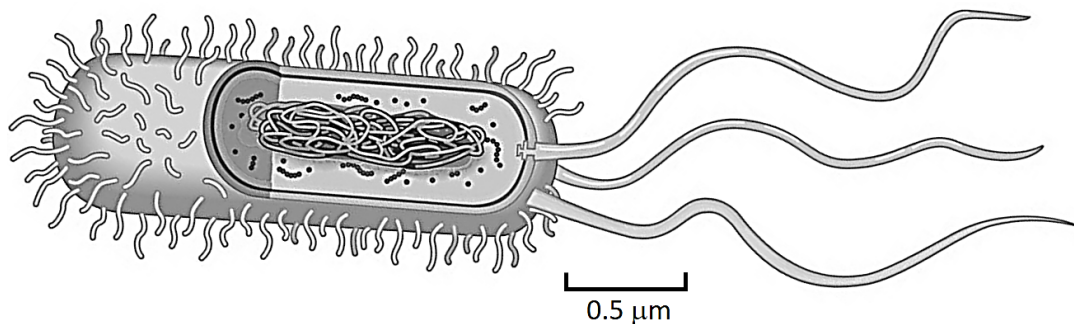


Fig. 1.4

- (i) With reference to Fig. 1.3 and Fig. 1.4, explain why Phylogenetic Tree A could appear to be unexpected. [3]

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- (ii) Although the domains Bacteria, Archaea, and Eukarya were founded based on genetic criteria, biochemical properties have also provided strong evidence that Archaea form an independent prokaryotic group and that they share features with both Bacteria and Eukarya. Major examples of these features are shown in Table 1.1.

Table 1.1

Features	Bacteria	Archaea	Eukarya
organization	unicellular	unicellular	unicellular and multicellular
nuclear envelope	absent	absent	present
DNA form	single, circular	single, circular	multiple, linear
histones associated with DNA	absent	present	present
nature of promoter	Pribnow box	TATA box	TATA box
presence of introns	no	in some genes	yes
types of RNA polymerase	Bacterial in nature	Eukarya-like	Eukaryal in nature
transcription factors	absent	present	present
operon	present	present	absent
initiator amino acid during translation	formyl-methionine	methionine	methionine
ribosomes	70S	Eukarya-like	80S
peptidoglycan in cell wall	present	absent	absent
response to the antibiotics streptomycin and chloramphenicol	inhibited	not inhibited	not inhibited

Explain which phylogenetic tree (**A** or **B**) is supported by the data in Table 1.1. [3]

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- (iii) One of the structural similarities shared between Archaea and Eukarya is the use of histone proteins in DNA packaging. Fig. 1.5 shows how DNA is packaged in eukaryotic and archaeal cells.

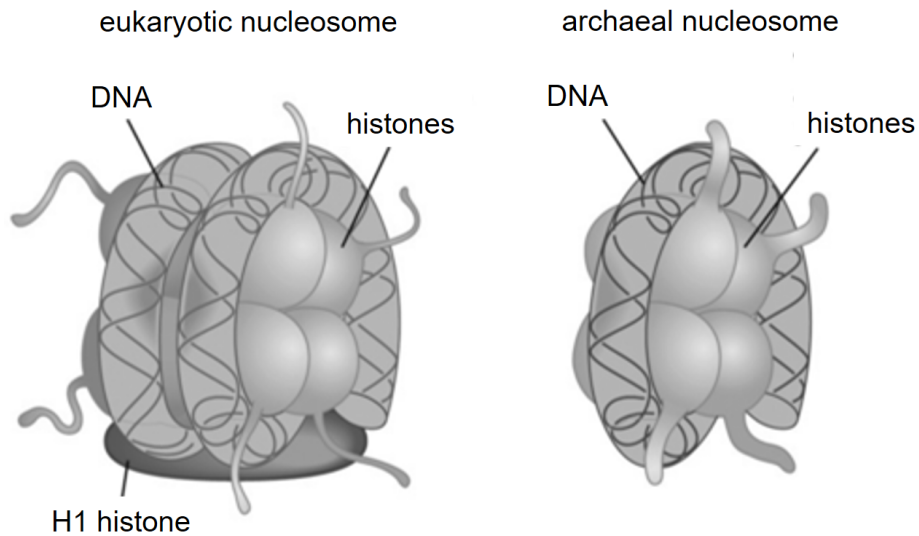


Fig. 1.5

State the differences in DNA packaging between eukaryotic and archaeal cells. [2]

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- (d) In eukaryotic cells, proteins destined for secretion are first synthesized by ribosomes bound to the rough endoplasmic reticulum (ER) membrane. The synthesized polypeptides then enter the ER lumen, where they are packaged into transport vesicles that undergo exocytosis. Exocytosis is not possible in archaeal and bacterial cells due to the lack of internal membranes.

Using the above information, suggest how archaeal and bacterial cells secrete proteins. [3]

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- (e) Although some Archaea inhabit the human body and possess some characteristics of human pathogens, surprisingly, no pathogenic archaeal species has been identified to date.

Characteristics possessed by human pathogens that are shared by Archaea include:

- They are highly diverse and are present in large numbers in the environment that would afford them the opportunity to cause disease.
 - They are recognized by antigen-presenting cells, B cells and phagocytes.
 - Some possess genes that code for toxin proteins needed for pathogenicity.
- (i) Some scientists have partly attributed the lack of pathogenic archaeal cells to the observation that there are far less viruses that infect Archaea than viruses that infect Bacteria.

Suggest why scientists have made this attribution. [2]

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- (ii) Using Table 1.1 on page 7, suggest why the emergence of pathogenic archaeal cells could threaten the entire human population. [1]

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- (ii) Suggest why the proposal to classify viruses into a fourth domain of life has been a subject of controversy. [3]

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[Total: 31]



QUESTION 2

Proteins must fold into defined three-dimensional structures to gain functional activity. In the cellular environment, newly synthesized polypeptides are at great risk of misfolding and aggregation. Cells hence engage proteins called chaperones to assist in protein folding.

These chaperones have two roles:

1. They bind to proteins to promote folding.
2. They direct misfolded polypeptides for degradation in the cytosol.

However, polypeptides that are in the midst of folding may be mistaken by chaperones as misfolded proteins and hence are directed for degradation. Therefore, protein folding needs to be completed quickly to prevent premature degradation.

A recently discovered rough endoplasmic reticulum (ER) protein complex called S-E complex was found to delay premature degrading of polypeptides that are in the midst of folding. In its absence, approximately 30% of newly synthesized proteins that could otherwise fold correctly are degraded.

Fig. 2.1 illustrates these processes.

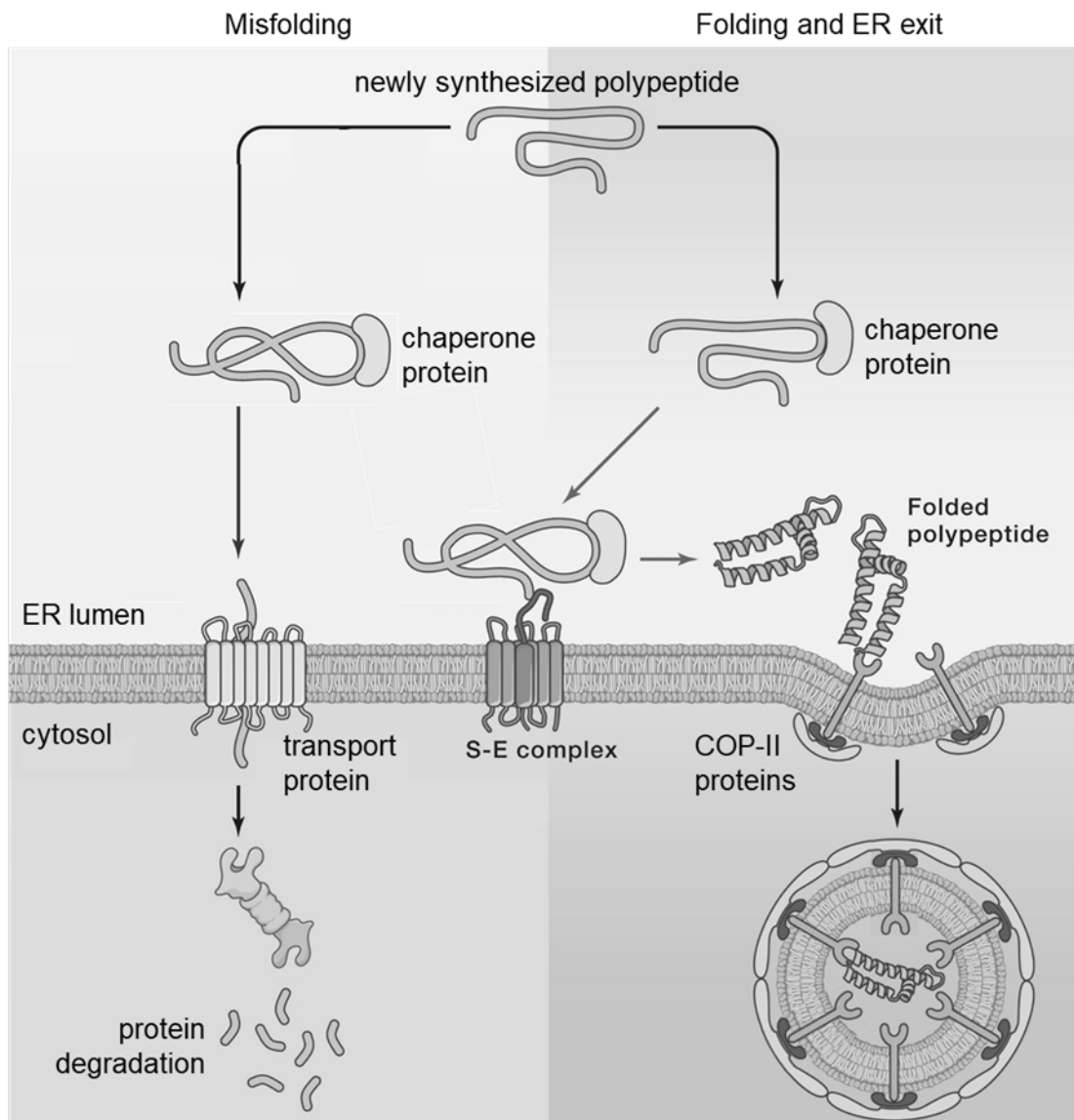


Fig. 2.1

- (a) Protein folding is driven mainly by the primary structure of the protein. In the midst of folding, the R-groups of polypeptides are exposed to their surrounding environment that contains many other different molecules.

Using the information above, suggest why it is important to have chaperone proteins to assist in protein folding. [1]

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- (b) With reference to Fig. 2.1, suggest how the S-E complex allows polypeptides to complete their folding. [2]

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- (c) ER vesicle formation is not a random event but is carefully orchestrated.

With reference to Fig. 2.1, describe how ER vesicle formation is triggered. [2]

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- (d) With reference to Fig. 2.1 and your knowledge on protein degradation, explain how unfolded polypeptides in the ER are degraded. [3]

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(e) Misfolded proteins in the ER tend to spontaneously associate with one another to form an aggregate, causing cellular toxicity. It is hence important that any misfolded protein is immediately degraded and not remain in the ER for too long.

(i) Explain why misfolded proteins in the ER tend to spontaneously associate with one another to form an aggregate. [2]

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(ii) Suggest why accumulation of protein aggregates in the ER is toxic to the cell. [1]

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- (f) Protein aggregates are usually due to genetic diseases. One classic example is Huntington's disease, a neurodegenerative disorder with a wide variation in the age of onset. A mutation in the *HTT* gene causes Huntington's disease. The *HTT* gene codes for a protein called huntingtin.

The *HTT* mutation that causes Huntington's disease involves a DNA segment in the *HTT* gene known as a CAG trinucleotide repeat. This segment is made up of a series of CAG that appears multiple times in a row.

An increase in the number of CAG repeats leads to the production of an abnormally long version of the huntingtin protein. The elongated protein is cut into smaller, toxic fragments that aggregate together and accumulate in neurons, disrupting the normal functions of these cells.

Fig. 2.2 shows how the number of CAG repeats varies in 1200 people with different ages of onset. Each data point represents an individual person.

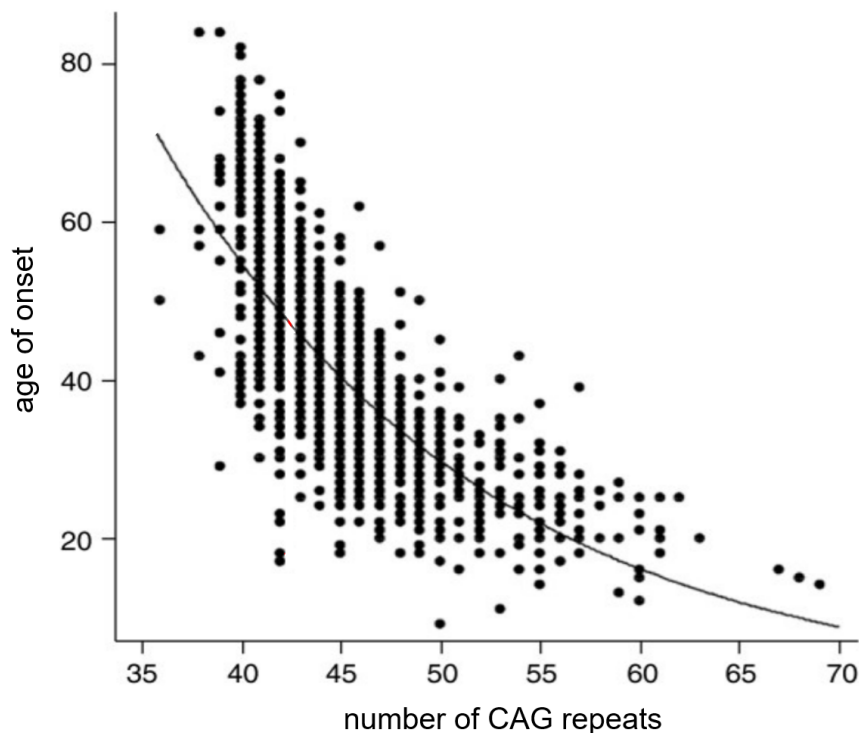


Fig. 2.2

- (i) Deduce if Huntington's disease is a dominant or recessive genetic disorder. [1]

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- (ii) With reference to Fig. 2.2, evaluate the extent to which the number of CAG repeats can predict the age of onset of Huntington's disease. [3]

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- (iii) Explain how molecular techniques can be used to estimate the number of CAG repeats in total DNA isolated from a person with Huntington's disease. [4]

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[Total: 19]



QUESTION 3

- Describe how tryptophan synthesis in bacterial cells is regulated **and** explain the advantages of such a regulation system in the context of lactose catabolism. [10]

[Total: 25]

QUESTION 4

- Explain how intramuscular injection with the mRNA vaccine leads to the protection against SARS-CoV-2 **and** discuss the factors that affect the extent of transmission of the disease. [15]

[Total: 25]

[illegible]

